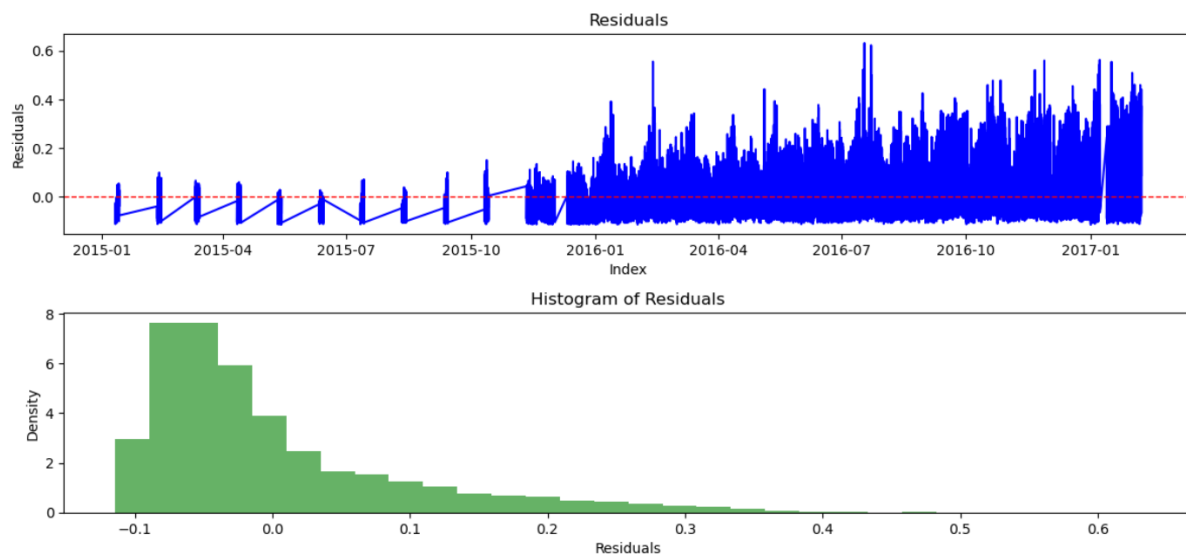


ARIMA MODEL

Model Evaluation

- **Mean Absolute Error (MAE):** 0.10326566194604825
- **Root Mean Square Error (RMSE):** 0.15309095534659473
- **R-squared:** -0.09108632425292007



Residual Analysis

- **Time Series Plot of Residuals:** The plot shows that the residuals are not randomly distributed around zero and have a clear pattern. This indicates that the model is not a good fit for the dataset.
- **Histogram of Residuals:** The histogram suggests that the residuals are not normally distributed and have a skewed distribution. This further supports the conclusion that the model is not a good fit for the dataset.

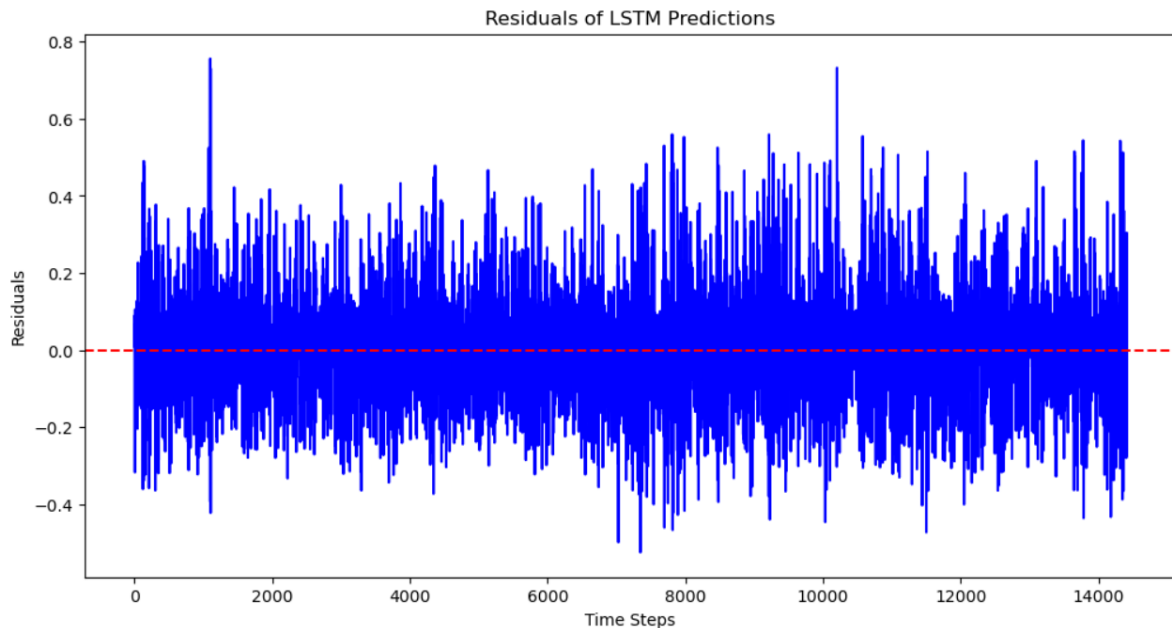
Conclusion

Based on the evaluation metrics and the residual analysis, the ARIMA model is not a good fit for the dataset. The model has a negative R-squared, which indicates that it does not explain the variation in the dataset. Additionally, the residuals are not randomly distributed or normally distributed, which indicates that the model has not captured the underlying patterns in the dataset.

LSTM MODEL

Evaluation Metrics:

- **MAE:** 0.0760
- **RMSE:** 0.1107
- **R-squared:** 0.4292



Residual Plot:

- The plot shows the residuals of the LSTM predictions.
- The residuals are centred around zero, indicating an unbiased model.
- High variability and some autocorrelation suggest the model is not capturing all patterns in the data.

Overall Assessment:

- The model performs moderately well with unbiased predictions but room for improvement.
- High variability and autocorrelation indicate potential for better model design.
- R-squared is relatively low, suggesting limited variance explanation.

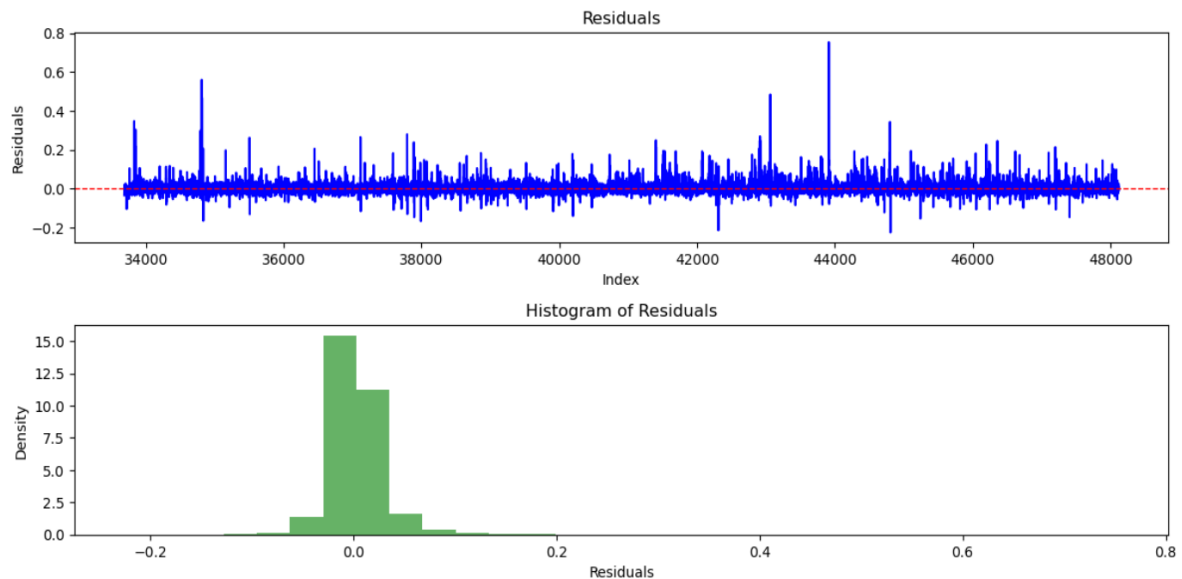
Conclusion:

The LSTM model provides a reasonable baseline but can be further improved for better accuracy and performance.

GRADIENT BOOSTING TREES

Model Evaluation Metrics:

- **MAE (Mean Absolute Error):** 0.018195356821334083
- **RMSE (Root Mean Squared Error):** 0.030153241861316554
- **R-squared:** 0.9576719688294908



Residual Analysis:

- **Residual Plot:** The residual plot shows that the residuals are randomly scattered around zero, indicating that the model is performing well and there is no significant pattern or trend in the errors.
- **Histogram of Residuals:** The histogram of residuals shows a slightly right-skewed distribution, indicating that the model may have a tendency to underestimate some values.

Interpretation:

The model exhibits strong performance with low MAE and RMSE values, indicating minimal prediction errors. The high R-squared value further supports this conclusion, suggesting that the model explains a significant portion of the variance in the data. The residual plot and histogram suggest that the model is generally performing well but may need some minor adjustments to address the slight right-skewness in the residuals.

Overall, the Gradient Boosting model is a good fit for the data and is producing accurate predictions.